

Daniele Battesimo Provenzano

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EDUCATION

Scuola Normale Superiore

Pisa, Italy

Ph.D. in Physics: Electromagnetics and Metamaterials;

Nov. 2020 – Sept. 2025

Diploma (M.Sc.) in Physics of Matter; 100/100 cum Laude

Oct. 2018 – Sept. 2020

University of Pisa

Pisa, Italy

M.Phil. in Condensed Matter Physics; 110/110 cum Laude

Sept. 2018 – Oct. 2020

Scuola Superiore di Catania

Catania, Italy

Diploma (M.Sc.) in Physics; 70/70 cum Laude

Oct. 2015 – Sept. 2018

University of Catania

Catania, Italy

B.Sc. in Physics; 110/110 cum Laude

Oct. 2015 – July 2018

ADDITIONAL COURSES

School for Forensic Police Inspectors: Selected through a national open competitive examination and completed a six-month training program in forensic science during Ph.D. studies. The program included Italian and international law, as well as experimental techniques used by the *Italian Scientific Police Department*, with a specialization in ballistics and gunshot residue analysis. Successfully appointed as *Technical Ballistic Inspector* upon completion of the training. (June 2023 - Gen. 2024)

“24 ECTS for Teaching”: Took four courses in Pedagogy, Psychology, Anthropology and Teaching Methods, which constitute a prerequisite for teaching in Italy. (Summer 2022)

TEACHING EXPERIENCE

Scuola Normale Superiore

Pisa, Italy

Lecturer, Teaching Assistant and Examiner

- **“Physics School in Pisa”:** Prepared and gave 28 hours of lectures about topics in Classical and Modern Physics and led the relative problem-solving sessions, examined and graded 200 applicants per year through both written and oral exams. (2019 - 2025)
- **“University Orientation Courses”:** Designed and delivered 12 hours of interactive lessons on Problem-Solving Techniques, aimed at 260 handpicked high school students. (June - July 2024)
- **“Classical Mechanics and Thermodynamics” course:** Served as a teaching assistant, delivering 30 hours of problem-solving sessions for first-year university students. (Fall 2020 - Spring 2021)

Italian Association for the Teaching of Physics (AIF)

Various locations across Italy

Lecturer, Tutor and Examiner

- **“One step away from IPhO”:** Prepared and gave 32 hours of lectures on Celestial Mechanics, Electromagnetism and Thermodynamics, aimed at the top 15 Italian students preparing to compete in the International Physics Olympiad. Selected the team members taking part in IPhO and EuPhO through theoretical and experimental examinations. (May 2019/23/24/25)
- **Physics Olympiad Summer School:** Led 18 hours of problem-solving sessions on Orbital Mechanics, Estimation Problems, Dimensional Analysis and Fluid Dynamics, aimed at 20 handpicked students taking part in the Physics Olympiad. (Aug. 2019 and 2024)

- **“Training camp for the Physics Olympiad”**: Prepared and gave three lectures about Electrostatics, Problem-Solving Techniques and Fluid Dynamics, aimed at 20 handpicked students taking part in the national finals of the Physics Olympiad. Led the relative problem-solving sessions, prepared and graded the final theoretical competition. (Mar. 2023/24/25)
- **“Training camp for the Math Olympiad”**: Led two problem-solving sessions focused on Combinatorics and Game Theory, aimed at 25 handpicked students taking part in the national finals of the Math Olympiad. (May 2016 and 2017)

PROJECTS RELATED TO PHYSICS EDUCATION

“Problem-Solving Techniques in Physics, with Original Problems and Debunked Myths”

- In my spare time, I am writing a textbook on problem-solving techniques aimed at students preparing for the Physics Olympiad and first-year university students. The book is specifically designed to debunk common myths and address misconceptions often found in traditional teaching and textbooks. I have written approximately 200 pages to date and aim to complete it by 2027.

Team Physics Competition | [Italian website](#)

- In 2023, after four years of development, my university colleagues and I launched the first national Physics Team Competition—an annual, multi-stage event for high school students across Italy. Around 600 teams participate annually. The top 20 teams advance to the in-person national finals in Senigallia, structured to resemble a sporting event. My primary responsibility involves designing, refining and selecting the theoretical problems used in the competition.

Physics School in Pisa | [Italian website](#)

- Since 2018, a team of physics students of Scuola Normale Superiore (SNS) has organized an annual one-week school for 24 top-performing Italian high school students. The program combines advanced preparation for the Physics Olympiad with an introduction to university-level physics through both classroom instruction and laboratory activities. In 2019, the project won the “Didactics of Physics” award from the Italian Physical Society. Since enrolling at SNS, I have served on the admissions committee (written and oral evaluations) and delivered lectures and problem-solving sessions on Fluid Dynamics, Celestial Mechanics, Special Relativity and Modern Physics across seven editions (2019–2025). Lecture materials, including recordings and PDF notes, are available on the website in Italian.

Italian Physics Olympiad | [Italian website](#)

- Organized since 1987 by the Italian Association for the Teaching of Physics (AIF), the Italian Physics Olympiad selects and trains the country’s top students for international competitions such as IPhO and EuPhO. I joined the national training team in 2019. Since then, students I have trained have earned 2 gold, 16 silver and 28 bronze medals. In recognition of my contributions, I was appointed to the national committee responsible for preparing theoretical and experimental problems and evaluating contestants at each stage of the competition.

CONFERENCE TALKS AND POSTERS

Metamaterials’2025, Amsterdam (NL), Sept. 2025

Graviton-Polaritons in FQH Systems Coupled to Optical Cavities (P)

GIREP-EPEC Conference, Leiden (NL), July 2025

When Assumptions Make the Difference: The Curious Case of the Mistreated Bernoulli’s Equation (T)

META 2024, Toyama (JP), July 2024

Photonic Modes in Gyrotropic-Hyperbolic Heterostructures (P)

PUBLICATIONS

Daniele B. Provenzano, *A comprehensive tutorial on Bernoulli's Equation*, in preparation.

Daniele B. Provenzano, *The incredible amount of physics behind the Heron's Fountain*, in preparation.

Daniele B. Provenzano, *Debunking Boyle's Self-Flowing Flask*, under review.

Stefania Carletti and Daniele B. Provenzano, *Le Olimpiadi Europee della Fisica*, chapter of the *Speciale Olimpiadi 2025*, published by Associazione dell'Insegnamento della Fisica (AIF). (Coming in 2026)

Daniele B. Provenzano, Andrea Stefanini, [Unblowing bubbles: understanding the physics of bubble deflation through a straw](#), Am. J. Phys. **93**, 797–805 (2025)

Daniele B. Provenzano, Giuseppe C. la Rocca, [Bulk and surface modes in a one-dimensional gyro-uniaxial photonic crystal](#), Phys. Rev. A **110**, 033514 (2024).

Daniele B. Provenzano, Giuseppe C. la Rocca, [Interface mode between gyroelectric and hyperbolic media](#), J. Opt. Soc. Am. B **40**, 172 (2023).

AWARDS & ACHIEVEMENTS

Italy Team Leader at EuPhO 2025: Coached and led the Italian team at the European Physics Olympiad, held in Sofia, Bulgaria. (June 2025)

Best student of the Technical Ballistics Inspector course: after six months of training, I ranked first in the final examinations among the ballistics inspector trainees and third overall among all the forensic science trainees, and was awarded a bronze medal. (Dec. 2023)

First Place in the National Open Competitive Exam for Technical Ballistic Inspectors: Ranked first in the decennial national exam for ballistic experts to join the Italian Police forensic department. (Apr. 2023)

Gold Medal – 53rd Rudolf Ortway International Competition in Physics. (Feb. 2023)

Absolute Winner – 50th Rudolf Ortway International Competition in Physics. (Dec. 2019)

SNS Scholarship: Awarded full scholarship (including room, board and monthly stipend) after passing the selection process to enroll at Scuola Normale Superiore as a Master's student. (Oct. 2018 – Sept. 2020)

SSC Scholarship: Granted full scholarship (room and board) after passing the selection process to enroll at Scuola Superiore di Catania as an undergraduate student. (Oct. 2015 – Sept. 2018)

CERTIFICATIONS AND SKILLS

Languages: English (C1, 195/210, certificated by Cambridge English), French (B1, certificated by Scuola Normale Superiore).

Major computer skills: MATLAB, Mathematica, Python, L^AT_EX, Git, HTML, video editing SWs.

Others: First Aid (practical knowledge, certificated by the Italian Police).

HOBBIES

DIY, Gardening, Plumbing, Football, Jogging, Bass guitar, 8- and 9-ball, Editing short movies.